Root Cause Analysis RCA-2022-0825

P-101 non-drive-end bearing seizure of 2022-08-20 | Issued 2022-08-25

RCA number	RCA-2022-0825	**Incident reference**	INC-2022-0820
Equipment	P-101, LP-101A	**Methodology**	Five-why with evidence verification
Team lead	Priya Sharma, Reliability Engineer	**Team**	Rajesh Kumar, D. Krishnan
Issued	2022-08-25	**Status**	Closed — corrective actions tracked to completion

1. Root cause statement

Lubrication failure. Internal wear in auxiliary lube oil pump LP-101A allowed the oil header pressure to decay below 1.4 bar, starving the P-101 non-drive-end bearing of oil film until the rolling elements welded to the outer race.

2. Five-why chain

#	Question	Answer
1	Why did P-101 stop?	The NDE bearing seized and the shaft locked.
2	Why did the bearing seize?	It ran without an oil film and the rolling elements welded to the race.
3	Why was there no oil film?	Lube oil header pressure fell to 1.2 bar, below the 1.4 bar minimum.
4	Why did header pressure fall?	LP-101A internal clearances had opened up from wear; volumetric efficiency dropped to roughly 55%.
5	Why was that wear not detected?	LP-101A carried no condition-monitoring task and lube oil analysis had lapsed for 18 months.

3. Evidence examined

- Failed bearing: rolling elements welded to the outer race at the load zone; cage fragmented. Consistent with oil-film loss, not with overload or misalignment.
- LP-101A strip-down: rotor and idler clearances opened to 0.31 mm against a 0.06 mm build figure. Volumetric efficiency calculated at approximately 55%.
- DCS trend: lube oil header pressure decayed from 2.1 bar to 1.2 bar over six days.
- Alarm history: two VT-1011 high-vibration alarms acknowledged on the night shift of 19 August 2022 with no work order raised.
- Oil analysis: last sample November 2021, 18 months before the event, against a 3-month standard interval.
- June 2022 inspection record: bearing wear 84% against the 85% OEM replacement limit, not escalated.

4. Contributing factors

1. LP-101A had no condition-based maintenance task; it was run to failure by default.
2. Lube oil analysis had not been performed since November 2021 (18-month interval against a 3-month standard).
3. The VT-1011 high-vibration alarm was acknowledged twice on the night shift of 19 August 2022 without a corrective work order being raised.
4. Bearing wear at 84% at the June 2022 inspection did not trigger the OEM 85% replacement rule because the trend was not plotted against the limit.

5. Corrective actions

#	Action	Owner	Target date
1	Overhaul LP-101A and restore rated discharge pressure of 2.1 bar	Rajesh Kumar	2022-08-24
2	Replace P-101 NDE and DE bearings per SOP-MECH-014	Rajesh Kumar	2022-08-26
3	Reinstate quarterly lube oil analysis for P-101 and P-102	Priya Sharma	2022-09-15
4	Add lube oil header low-pressure alarm PI-1016 at 1.4 bar to the DCS alarm list	D. Krishnan	2022-09-30
5	Plot bearing wear against the 85% OEM limit on every inspection report	Priya Sharma	2022-10-14

6. Failure mechanism, in plain language

The auxiliary lube oil pump wore out quietly. Nothing watched it, because it had no condition-monitoring task and no low-pressure alarm. As its internal clearances opened up, the oil header pressure fell a little each week. Below 1.4 bar the oil film in the non-drive-end bearing could no longer separate the rolling elements from the race. The bearing ran metal-to-metal, heated, and welded itself solid. The vibration alarms that should have interrupted this were acknowledged twice on a night shift and no work order was raised.

> *The failure did not begin on 20 August 2022. It began when a 3-month oil analysis interval was allowed to run for 18 months, and it became inevitable when two alarms were acknowledged without action.*

7. Effectiveness review

Action	Completed	Evidence	Effective
LP-101A overhaul	2022-08-24	Work order MR-2022-0824	Yes — 2.1 bar restored
Bearing replacement	2022-08-26	Work order MR-2022-0826	Yes — post-job vibration 3.1 mm/s
Quarterly oil analysis reinstated	2022-09-15	Lab schedule LS-2022-09	Partial — lapsed again in 2023
PI-1016 low-pressure alarm added	2022-09-30	DCS alarm list Rev 12	Yes
Wear plotted against the OEM limit	2022-10-14	Inspection template Rev 3	Partial — not applied on every report

Two corrective actions are recorded as only partially effective. Both concern the detection of exactly the condition now present on this machine.